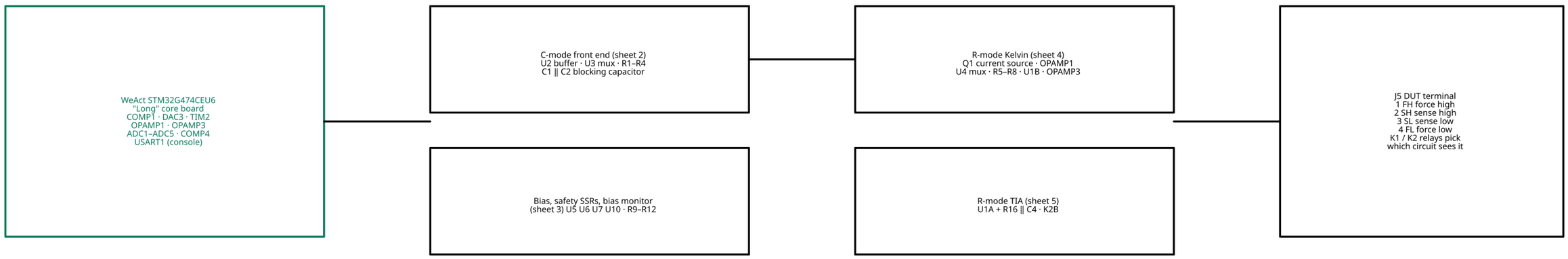
How the board is organised



Power, relay drivers, console, J8 header, full pin map: sheet 6 · complete KiCad schematic (every wire): sheet 7

Measurement modes (set by firmware through PB9 → K1, PB10 → K2)

Mode	K1 (R_MODE)	K2 (KELVIN)	DUT bottom FL goes to	DUT top FH driven by	Range
C	off (NC)	off (NC)	GND via K1A	C_BLOCK via K1B	pF – mF
Kelvin R	on (NO)	on (NO)	MUXR_COM → U4 → R5–R8	Q1 current source	≈0.1 Ω – 22 MΩ
TIA R	on (NO)	off (NC)	U1A summing node	Q1 fully on	≈15 MΩ – 1 GΩ

Power-off / reset state = C-mode (both relays de-energised: FL grounded, C_BLOCK connected).

Every amplifier / comparator in the design

Device	Part / location	Job	Sheet
U1A	LMC6482AIM/NOPB (TI), external SOIC-8	TIA: converts the nA DUT current to a voltage (R16 10 MΩ)	5
U1B	LMC6482AIM/NOPB (TI), external SOIC-8	Unity buffer for SL (DUT sense-low) → VBUF	4, 5
OPAMP1	STM32G474 internal op-amp	Kelvin current loop: drives Q1 so MUXR_COM = VREF (0.405 V)	4
OPAMP3	STM32G474 internal op-amp	V_SH buffer; Kelvin: PGA ×2…×64 referenced to VBUF	3, 4, 5
COMP1	STM32G474 internal comparator	C-mode relaxation oscillator (70 mV hysteresis)	2
COMP4	STM32G474 internal comparator	R-mode V_N over-voltage guard (2.6 V)	4
DAC3_CH1 / DAC1_CH1	STM32G474 internal DACs (no pin)	COMP1 threshold 1.65 V / COMP4 threshold 2.6 V	2, 4

TLC2272 / TLC2262 are not used: they need ≥ 4.4 V and the carrier runs on 3.3 V.

Firmware note: OPAMP3 PGA mode, PC4 (U10), PB15 (MUXR_EN) and PB2 (VBUF) follow docs/accuracy-plan.md F1–F3 (firmware update pending).

Legend:

blue text = net name (identical to the KiCad schematic)

green dashed box = inside the STM32; ◦ PAx (J2-n) = MCU pin (WeAct header pin)

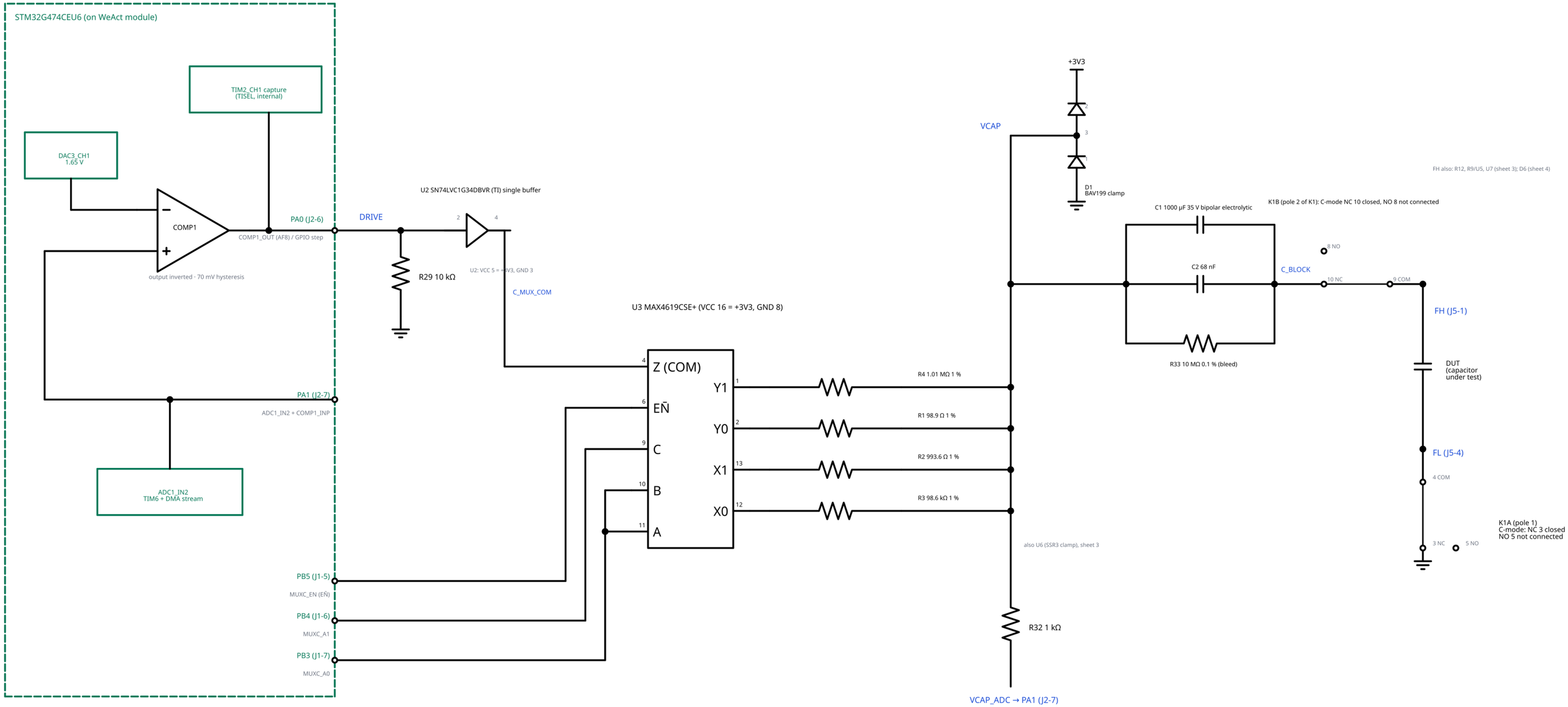
small numbers next to a part = its pin numbers

ESWCap C+R Meter — carrier board Rev C.1

Sheet 1 — System overview, modes and amplifier list

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Netlist-verified against hardware/kicad/Project.kicad_sch



U3 truth table (A = B = MUXC_A0 on PB3, C = MUXC_A1 on PB4): 00 → X0 R3 98.6 kΩ 01 → X1 R2 993.6 Ω 10 → Y0 R1 98.9 Ω 11 → Y1 R4 1.01 MΩ

MUXC_EN (PB5) has R30 100 kΩ pull-up to +3V3: U3 is off at reset (EN̄ active low).

The DUT is in series with C1 ‖ C2 (C_block ≈ 1000 µF): C_DUT = C_eq·C_block / (C_block − C_eq).

Oscillator: COMP1 drives U2 through PA0, T = K·R·C + T0 (K ≈ 0.085), timed by TIM2. RC step: PA0 GPIO step, ADC1 records V(t), τ = R·C.

Legend:

blue text = net name (identical to the KiCad schematic)

green dashed box = inside the STM32; ◦ PAx (J2-n) = MCU pin (WeAct header pin)

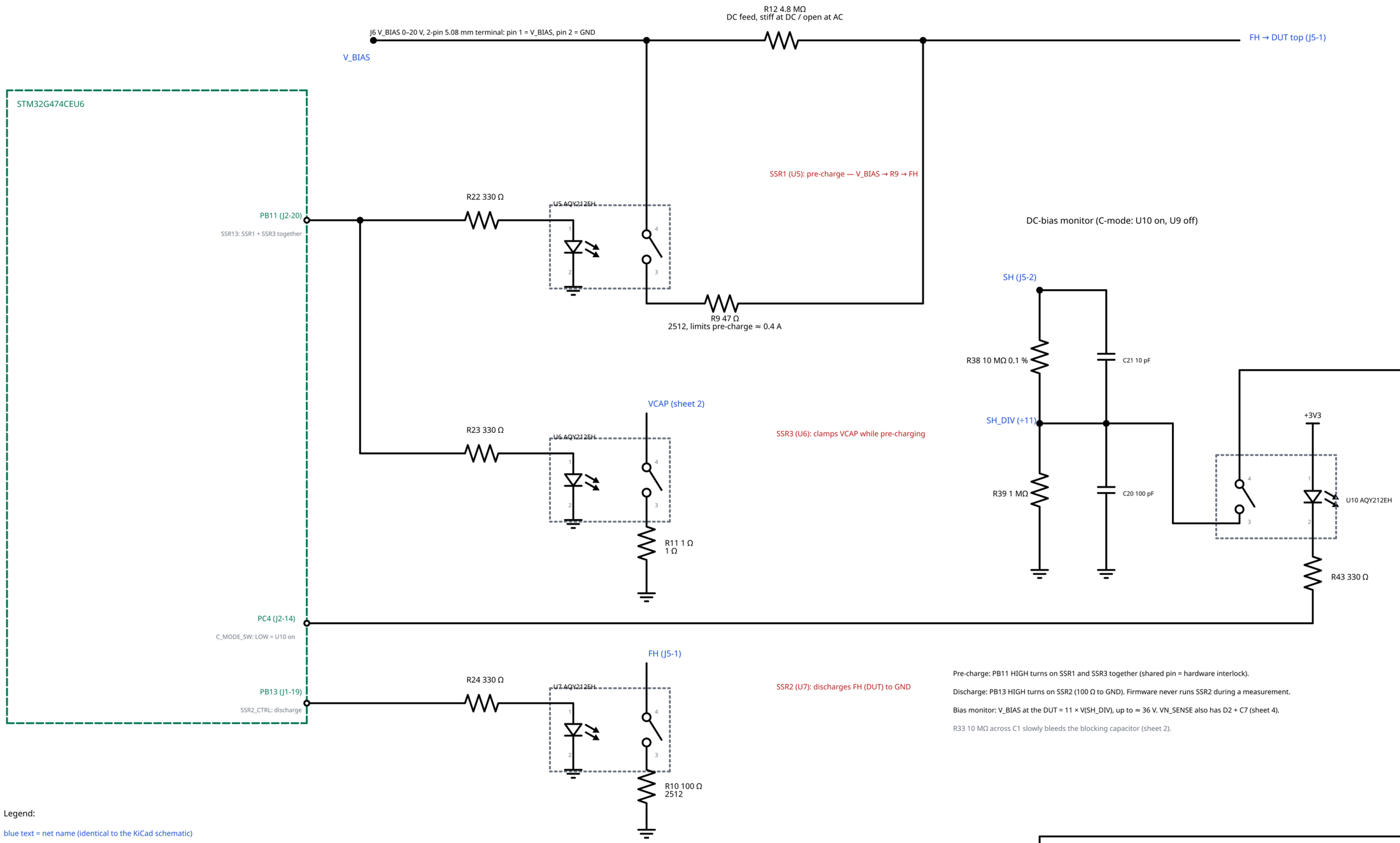
small numbers next to a part = its pin numbers

ESWCap C+R Meter — carrier board Rev C.1

Sheet 2 — C-mode measurement path (K1 and K2 de-energised)

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Netlist-verified against hardware/kicad/Project.kicad_sch



Pre-charge: PB11 HIGH turns on SSR1 and SSR3 together (shared pin = hardware interlock).

Discharge: PB13 HIGH turns on SSR2 (100 Ω to GND). Firmware never runs SSR2 during a measurement.

Bias monitor: V_BIAS at the DUT = 11 × V(SH_DIV), up to ≈ 36 V. VN_SENSE also has D2 + C7 (sheet 4).

R33 10 MΩ across C1 slowly bleeds the blocking capacitor (sheet 2).

Legend:

blue text = net name (identical to the KiCad schematic)

green dashed box = inside the STM32; ◦ PAx (J2-n) = MCU pin (WeAct header pin)

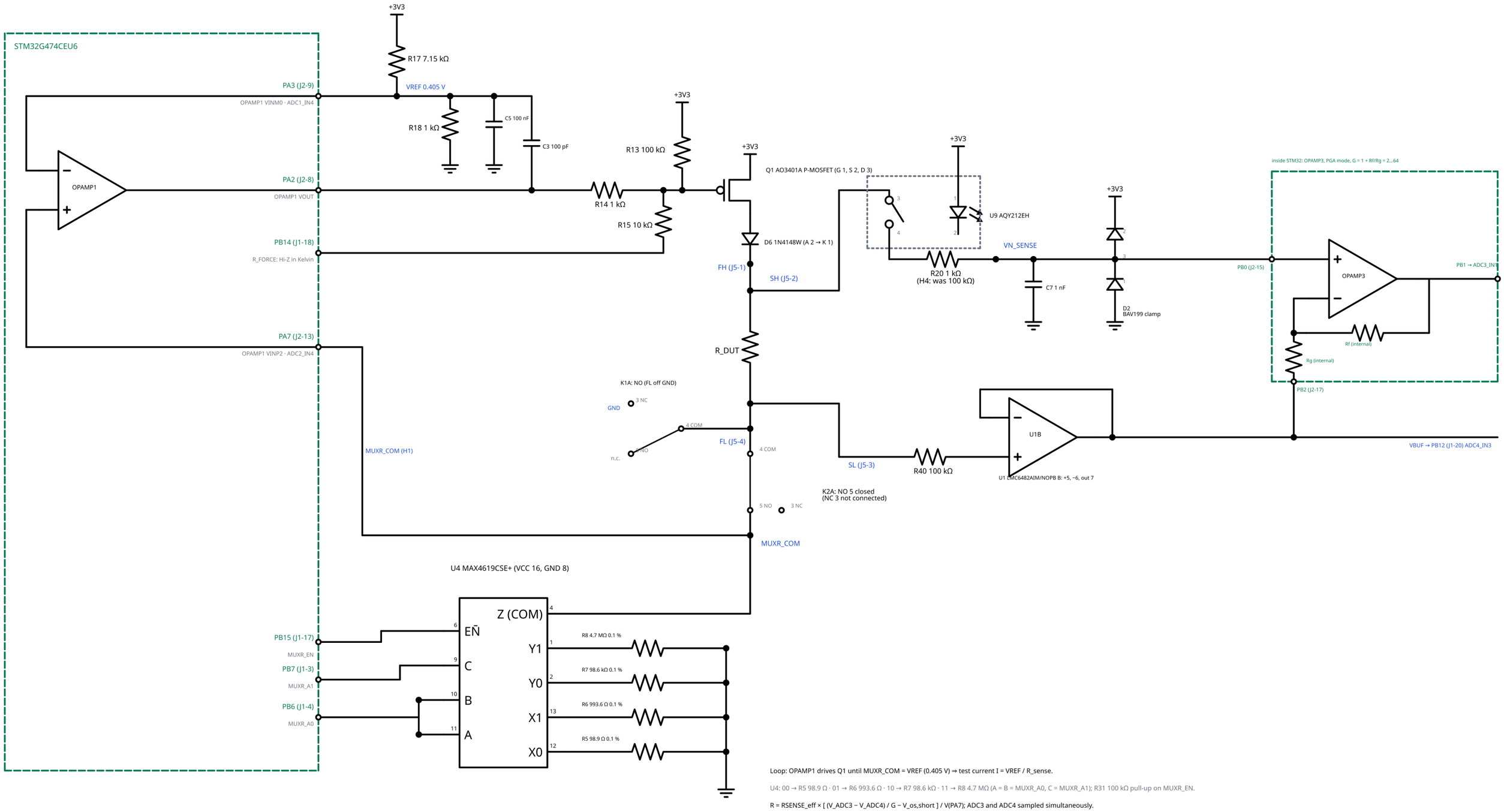
small numbers next to a part = its pin numbers

ESWCap C+R Meter — carrier board Rev C.1

Sheet 3 — DUT bias, safety SSRs and DC-bias monitor (C-mode)

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Netlist-verified against hardware/kicad/Project.kicad_sch



Loop: OPAMP1 drives Q1 until MUXR_COM = VREF (0.405 V) ⇒ test current I = VREF / R_sense.

U4: 00 → R5 98.9 Ω · 01 → R6 993.6 Ω · 10 → R7 98.6 kΩ · 11 → R8 4.7 MΩ (A = B = MUXR_A0, C = MUXR_A1); R31 100 kΩ pull-up on MUXR_EN.

R = RSENSE_eff × [(V_ADC3 − V_ADC4) / G − V_os,short] / V(PA7); ADC3 and ADC4 sampled simultaneously.

COMP4 (internal): PB0 vs DAC1_CH1 2.6 V — over-voltage guard shuts R-mode down. OPAMP3 out = V_SL + G·(V_SH − V_SL).

U9 LED: pin 1 = +3V3, pin 2 → R41 330 Ω → K1_C, so U9 conducts whenever K1 is energised (R-mode only).

Legend:

blue text = net name (identical to the KiCad schematic)

green dashed box = inside the STM32; ◦ PAx (J2-n) = MCU pin (WeAct header pin)

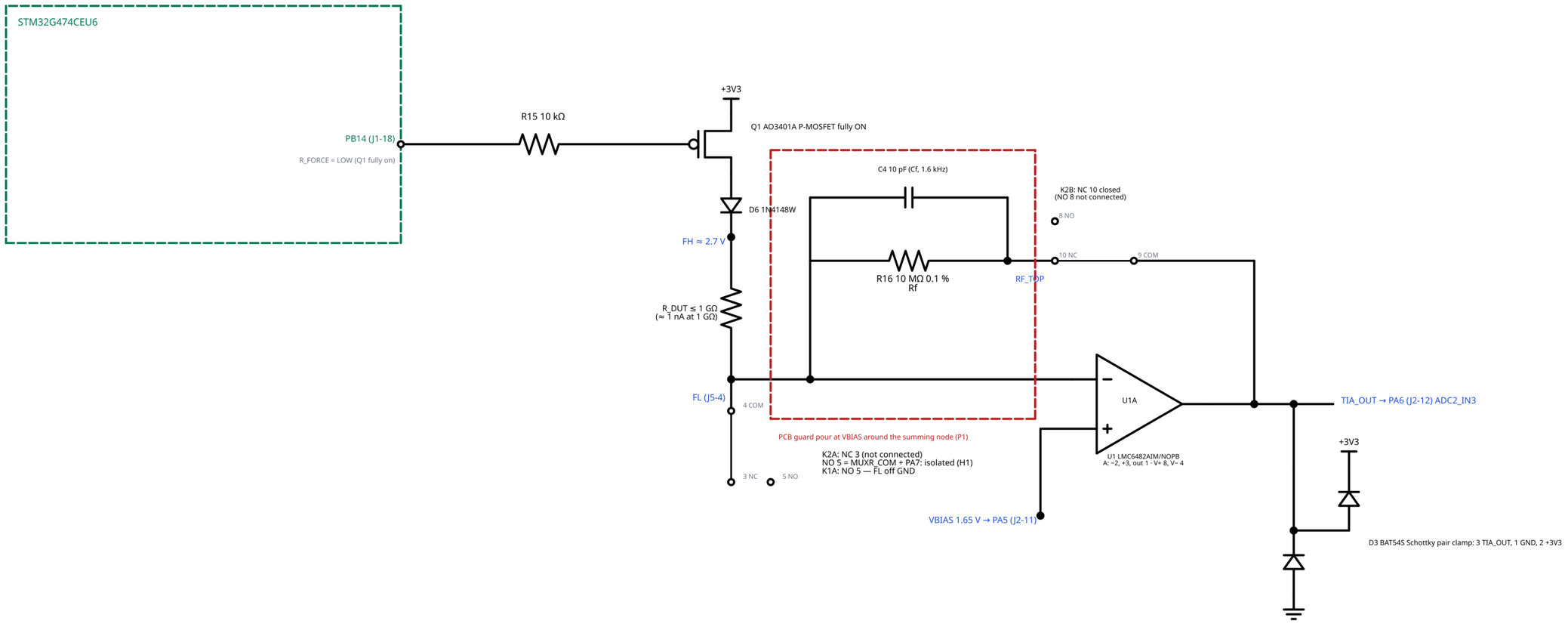
small numbers next to a part = its pin numbers

ESWCap C+R Meter — carrier board Rev C.1

Sheet 4 — R-mode, 4-wire Kelvin (K1 energised, K2 energised)

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Netlist-verified against hardware/kicad/Project.kicad_sch



OPAMP1 is off (PA2 Hi-Z); R13 and R14 on the gate (sheet 4) do not matter here. PB14 LOW pulls the gate to ≈ 0.3 V.

V_SH: SH → U9 → R20 → OPAMP3 (follower) → PB1 → ADC3. V_SL: SL → R40 → U1B → VBUF → PB12 → ADC4 (sheet 4).

VBIAS = 1.65 V from R19 10 kΩ (+3V3 → VBIAS) and R21 10 kΩ (VBIAS → GND), C6 100 nF; read on PA5 (J2-11) ADC2_IN13.

U1A holds FL at VBIAS and turns the DUT current into a voltage across Rf: V_TIA = V_FL − I × Rf.

I = (V_SL − V_TIA − V_open) / Rf R = Rf × (V_SH − V_SL) / (V_SL − V_TIA − V_open) (V_open from the open-circuit tare)

Legend:

blue text = net name (identical to the KiCad schematic)

green dashed box = inside the STM32; ◦ PAx (J2-n) = MCU pin (WeAct header pin)

small numbers next to a part = its pin numbers

ESWCap C+R Meter — carrier board Rev C.1

Sheet 5 — R-mode, TIA for 15 MΩ – 1 GΩ (K1 energised, K2 de-energised)

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Netlist-verified against hardware/kicad/Project.kicad_sch

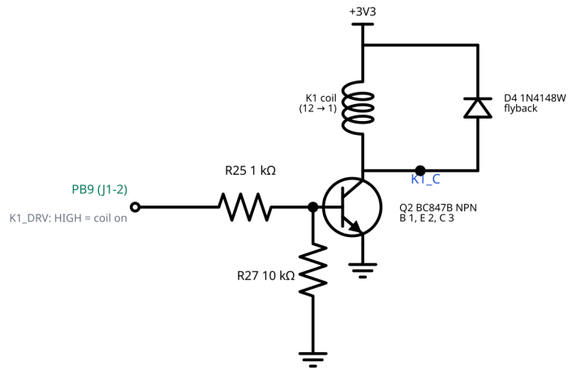
Power (USB only)

USB-C VBUS → B5819WS → VDD5V → ME6231A33M3G LDO → +3V3 (all on the WeAct module)

J4-3/4 = +3V3 to carrier, J4-1/2 = GND; J2-2 = +3V3; J1-1, J2-1 = GND; J3-1/2 = GND, J3-3/4 (VDD5V) not connected.

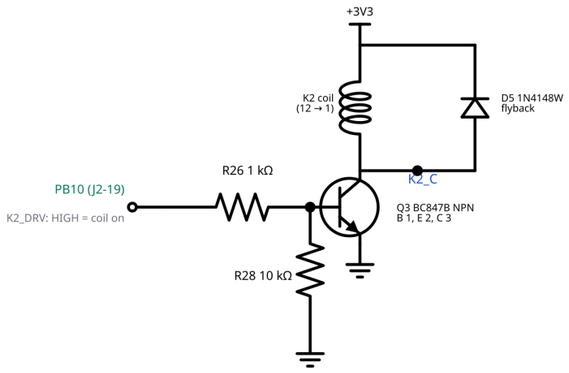
Decoupling: C8 100 nF + C9 10 µF (bulk), C10 100 nF + C15 10 µF (relay coils), C11–C14 100 nF (one per IC: U1–U4).

K1 driver



K1_C also drives U9 LED through R41 (sheet 4). Firmware never energises K2 without K1.

K2 driver



Console J7 (115200 8N1)

1 UART_TX ← PA9 (J1-14) USART1_TX
2 UART_RX → PA10 (J1-13) USART1_RX
3 GND 4 +3V3

Bias-control header J8 (future bias module)

1 BIAS_SET_F ← R34 1 kΩ ← BIAS_SET = PA4 (J2-10) TIM3_CH2 PWM; C16 10 µF to GND
2 BIAS_EN = PA15 (J1-10) GPIO; R35 10 kΩ pull-down
3 BIAS_MON_F → R36 1 kΩ → BIAS_MON = PA8 (J1-15) ADC5_IN1; C17 100 nF to GND
4 GND 5 +3V3 (< 50 mA) 6 BIAS_UART_TX = PC10 (J1-9) 7 BIAS_UART_RX = PC11 (J1-8) 8 GND

Complete MCU pin map (WeAct header pin → STM32 pin → net → peripheral)

Header	MCU	Net	Peripheral	Mode	Header	MCU	Net	Peripheral	Mode
J2-6	PA0	DRIVE	COMP1_OUT (AF8) / GPIO	C	J1-3	PB7	MUXR_A1	GPIO → U4 C	Kelvin
J2-7	PA1	VCAP_ADC	ADC1_IN2, COMP1_INP	C	J1-4	PB6	MUXR_A0	GPIO → U4 A, B	Kelvin
J2-8	PA2	OP1_OUT	OPAMP1_VOUT	Kelvin	J1-5	PB5	MUXC_EN	GPIO → U3 EN̄	C
J2-9	PA3	VREF	OPAMP1_VINM0, ADC1_IN4	Kelvin	J1-6	PB4	MUXC_A1	GPIO → U3 C	C
J2-10	PA4	BIAS_SET	TIM3_CH2 PWM	J8	J1-7	PB3	MUXC_A0	GPIO → U3 A, B	C
J2-11	PA5	VBIAS	ADC2_IN13	TIA	J1-8	PC11	BIAS_UART_RX	USART3_RX	J8
J2-12	PA6	TIA_OUT	ADC2_IN3	TIA	J1-9	PC10	BIAS_UART_TX	USART3_TX	J8
J2-13	PA7	MUXR_COM	OPAMP1_VINP2, ADC2_IN4	Kelvin	J1-10	PA15	BIAS_EN	GPIO	J8
J2-14	PC4	C_MODE_SW	GPIO (U10, active low)	C	J1-13	PA10	UART_RX	USART1_RX	console
J2-15	PB0	VN_SENSE	OPAMP3_VINP0, COMP4_INP	all	J1-14	PA9	UART_TX	USART1_TX	console
J2-16	PB1	(not on carrier)	OPAMP3_VOUT → ADC3_IN1	all	J1-15	PA8	BIAS_MON	ADC5_IN1	J8
J2-17	PB2	VBUF	OPAMP3_VINM0 (PGA bias)	Kelvin	J1-17	PB15	MUXR_EN	GPIO → U4 EN̄ (H2)	Kelvin
J2-19	PB10	K2_DRV	GPIO → Q3 → K2	R	J1-18	PB14	R_FORCE	GPIO (Hi-Z / LOW)	R
J2-20	PB11	SSR13	GPIO → U5 + U6	C	J1-19	PB13	SSR2_CTRL	GPIO → U7	C
J1-2	PB9	K1_DRV	GPIO → Q2 → K1	R	J1-20	PB12	VBUF	ADC4_IN3	R

Reserved by the module: PA11/PA12 USB, PA13/PA14 SWD, PF0/PF1 HSE 8 MHz, PC6 user LED, PC13 user key (active high), PB8 BOOT0, VREF+ = VDDA (R7 0 Ω).

Legend:

blue text = net name (identical to the KiCad schematic)

green dashed box = inside the STM32; ◦ PAx (J2-n) = MCU pin (WeAct header pin)

small numbers next to a part = its pin numbers

ESWCap C+R Meter — carrier board Rev C.1

Sheet 6 — Power, relay drivers, console, J8 bias header, MCU pin map

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Netlist-verified against hardware/kicad/Project.kicad_sch

ESWCap combined C+R front end (STM32G474 carrier)

REV C: USB-only; 4-wire C/R; OPAMP3 SH buffer; J5 FH/SH/SL/FL; PA0=DRIVE, PA6=BIAS_MON

C: DRIVE -> C mux -> Rsense -> Vcap -> C1/C2 -> K1B -> N_DUT -> DUT -> B -> K1A -> GND

REV B: C1 = 1000 uF / 35 V BIPOLAR (+ marking not used).

D6 (1N4148) blocks Q1 body diode; R9 = 47R 2W; C3 returns to VREF.

Console J7: PA9=TX, PA10=RX, GND, +3V3. See docs/pcb-design.md.

R: +3V3 -> Q1 -> N_DUT -> DUT -> B -> K2A -> R mux -> RSENSE -> GND. OPAMP1: VINP=B(PA7), VINM=VREF(PA3), VOUT=PA2. TIA: LMC6482 A.

J2
1 GND
2 +3V3
3
4
5
6 DRIVE
7 VCAP_ADC
8 OP1_OUT
9 VREF
10 BIAS_SET
11 VBIAS
12 TIA_OUT
13 MUXR_COM
14 C_MODE_SW
15 VN_SENSE
16
17 VBUF
18
19 K2_DRV
20 SSR13
WeAct P2

J1
1 GND
2 K1_DRV
3 MUXR_A1
4 MUXR_A0
5 MUXC_EN
6 MUXC_A1
7 MUXC_A0
8 BIAS_UART_RX
9 BIAS_UART_TX
10 BIAS_EN
11
12
13 UART_RX
14 UART_TX
15 BIAS_MON
16
17 MUXR_EN
18 R_FORCE
19 SSR2_CTRL
20 VBUF
WeAct P1

J3
1 GND
2 GND
3
4
WeAct P3 (NC)

J4
1 GND
2 GND
3 +3V3
4 +3V3
WeAct P4 3V3

J7
1 UART_TX
2 UART_RX
3 GND
4 +3V3
Console

U2
74LVC1G34
+3V3
DRIVE 2
4 C_MUX_COM
3 GND
5
1

U3
MAX4619
C_CH3 1 Y1
C_CH2 2 Y0
MUXC_Q 3 Z1
C_MUX_COM 4 Z
MUXC_P 5 Z0
MUXC_EN 6 ENABLE
7 NC
GND 8 GND
VCC 16 +3V3
Y 15 MUXC_Q
X 14 MUXC_P
X1 13 C_CH1
X0 12 C_CH0
A 11 MUXC_A0
B 10 MUXC_A0
C 9 MUXC_A1

U4
MAX4619
R_CH3 1 Y1
R_CH2 2 Y0
MUXR_Q 3 Z1
MUXR_COM 4 Z
MUXR_P 5 Z0
MUXR_EN 6 ENABLE
7 NC
GND 8 GND
VCC 16 +3V3
Y 15 MUXR_Q
X 14 MUXR_P
X1 13 R_CH1
X0 12 R_CH0
A 11 MUXR_A0
B 10 MUXR_A0
C 9 MUXR_A1

U5
AQY212
SSR1_LED 1 LED+
GND 2 LED-
SW 3 SSR1_SW
SW 4 V_BIAS

U6
AQY212
SSR3_LED 1 LED+
GND 2 LED-
SW 3 SSR3_SW
SW 4 VCAP

U7
AQY212
SSR2_LED 1 LED+
GND 2 LED-
SW 3 SSR2_SW
SW 4 FH

U9
AO3401
Q1_D 1 +3V3
2 U9_LED_K
LED+ 11 SW 3 SH
LED- SW 4 VN_IN

U10
AQY212
1 +3V3 LED+
2 U10_LED_K LED-
SW 3 SH_DIV
SW 4 VN_SENSE

K1
NC1 3 GND
COM1 4 FL
NO1 5 C_BLOCK
NC2 10 FH
COM2 9
NO2 8
COIL+ +3V3
COIL- 12 K1_C

K2
NC1 3
COM1 4 FL
NO1 5 MUXR_COM
NC2 10 RF_TOP
COM2 9 TIA_OUT
NO2 8
COIL+ +3V3
COIL- 12 K2_C
KELVIN

U1A
LMC6482
IN+ 3 VBIAS
IN- 2 FL
OUT 1 TIA_OUT

U1B
LMC6482
IN+ 5 VBUF
IN- 6
OUT 7 VBUF

U1C
LMC6482
V+ 8 +3V3
V- 4 GND

Q1
BC847
G Q1_GATE
S +3V3
D Q1_D

Q2
BC847
B K1_BASE
C K1_C
E GND

Q3
BC847
B K2_BASE
C K2_C
E GND

D1
BAV199
A1 GND
K2 +3V3
COM VCAP

D2
BAV199
A1 GND
K2 +3V3
COM VN_SENSE

D3
BAT54S
A1 GND
K2 +3V3
COM TIA_OUT

D4
1N4148
K1_C +3V3

D5
1N4148
K2_C +3V3

D6
1N4148
VCAP C_BLOCK
FH

J5
DUT
1 FH
2 SH
3 SL
4 FL

J6
V_BIAS 0–20V
1 V_BIAS
2 GND

J8
Pin_1 1 BIAS_SET_F
Pin_2 2 BIAS_EN
Pin_3 3 BIAS_MON_F
Pin_4 4 GND
Pin_5 5 +3V3
Pin_6 6 BIAS_UART_TX
Pin_7 7 BIAS_UART_RX
Pin_8 8 GND

R1 C_CH2 1 2 VCAP 98.9 1%
R2 C_CH1 1 2 VCAP 993.6 1%
R3 C_CH0 1 2 VCAP 98.6k 1%
R4 C_CH3 1 2 VCAP 1.01M 1%
R5 R_CH0 1 2 GND 98.9 0.1%
R6 R_CH1 1 2 GND 993.6 0.1%
R7 R_CH2 1 2 GND 98.6k 0.1%
R8 R_CH3 1 2 GND 4.7M 0.1%
R9 SSR1_SW 1 2 FH 47
R10 SSR2_SW 1 2 GND 100
R11 SSR3_SW 1 2 GND 1
R12 V_BIAS 1 2 FH 4.8M
R13 +3V3 1 2 Q1_GATE 100k
R14 OP1_OUT 1 2 Q1_GATE Q1_GATE 1k
R15 R_FORCE 1 2 Q1_GATE 10k
R16 FL 1 2 RF_TOP 10M 0.1%
R17 +3V3 1 2 VREF 7.15k
R18 VREF 1 2 GND 1k
R19 +3V3 1 2 VBIAS 10k
R20 VN_IN 1 2 VN_SENSE 1k
R21 VBIAS 1 2 GND 10k
R22 SSR13 1 2 SSR1_LED 330
R23 SSR13 1 2 SSR3_LED 330
R24 SSR2_CTRL 1 2 SSR2_LED 330
R25 K1_DRV 1 2 K1_BASE 1k
R26 K2_DRV 1 2 K2_BASE 1k
R27 K1_BASE 1 2 GND 10k
R28 K2_BASE 1 2 GND 10k
R29 GND 1 2 R_MODE 10k
R30 R_MODE 1 2 +3V3 100k
R31 +3V3 1 2 VCAP 100k
R32 VCAP 1 2 VCAP_ADC 1k
R33 VCAP 1 2 C_BLOCK 10M 0.1%
R34 BIAS_SET BIAS_SET_F 1k
R35 2 GND 1 BIAS_EN 10k
R36 BIAS_MON BIAS_MON_F 1k
R38 SH 2 SH_DIV 10M 0.1%
R39 SH_DIV 2 GND 1M
R43 C_MODE_SW U10_LED_K 330

C1 VCAP 1 2 C_BLOCK 1000uF/35V BP
C2 VCAP 1 2 C_BLOCK 68nF
C3 OP1_OUT 1 2 VREF 100pF
C4 FL 1 2 RF_TOP 10pF
C5 VREF 1 2 GND 100nF
C6 VBIAS 1 2 GND 100nF
C7 VN_SENSE 1 2 GND 1nF
C8 +3V3 1 2 GND 100nF
C9 +3V3 1 2 GND 10uF
C10 +3V3 2 GND 100nF
C11 +3V3 1 2 GND 100nF
C12 +3V3 1 2 GND 100nF
C13 +3V3 1 2 GND 100nF
C14 +3V3 1 2 GND 100nF
C15 +3V3 2 GND DRIVE 10uF
C16 BIAS_SET BIAS_SET_F 1k
C17 BIAS_MON BIAS_MON_F 10nF
C20 SH_DIV 2 GND 100pF
C21 SH 2 SH_DIV 10pF

FLG1 +3V3
FLG3 GND
FLG4 V_BIAS

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Sheet: /
File: Project.kicad_sch
Title: ESWCap C+R Meter — carrier board
Size: A3 | Date: 2026-11-14 | Rev: C.1
KiCad E.D.A. 10.0.8 | Id: 1/1